Prime Factors and Integer Exponents

Concept

Prime factors reveal shared structure. Integer exponents include zero and negatives.

GCF / LCM

Use smaller shared prime powers / larger powers of all primes.

Negative power

$a^{-n} = 1/a^n$ for $a \neq 0$; it does not mean a negative answer.

Laws

Add exponents in products; subtract in quotients; multiply in powers.

Scientific notation

$c \times 10^n$, with $1 \leq |c| < 10$.

Example 1

Find GCF and LCM of 72 and 120.

Solution

1. $72 = 2^3 \times 3^2$; $120 = 2^3 \times 3 \times 5$.
2. $GCF = 2^3 \times 3 = 24$.
3. $LCM = 2^3 \times 3^2 \times 5 = 360$.

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Example 2

Simplify $(3x^{-2}y^{3})/(6xy^{-1})$.

Solution

1. Original restrictions: $x \neq 0$, $y \neq 0$.

2. Combine: $(1/2)x^{-3}y^{4} = y^{4}/(2x^{3})$.

3. Keep both original restrictions.

Prime Factors and Integer Exponents

Practice | your turn

Show your steps. Include units and restrictions where needed.

1. Find GCF and LCM of 84 and 126.

2. Simplify 5^{-2}.

3. Simplify $(4x^3)/(2x^{-2})$, including restrictions.

4. Write 0.000072 in scientific notation.

5. Evaluate $(-2)^{-3}$.

Prime Factors and Integer Exponents

Practice answers | check after trying

1. $84 = 2^2 \times 3 \times 7$; $126 = 2 \times 3^2 \times 7$. GCF $= 42$ and LCM $= 252$.

2. $1/25$.

3. $2x^5$, $x \neq 0$.

4. 7.2×10^{-5}.

5. $1/(-2)^3 = -1/8$.